YASHWANTRAO CHAVAN MAHARASHTRA OPEN UNIVERSITY, NASHIK

(Estd. By Government of Maharashtra)
Dnyangangotri, Near Gangapur Dam, Govardhan, Nashik-422 222
Home Assignment

Academic Year - 2025 - 2026

Name of the Programme & Prog. Code	P131 - B.C.A.(with Credits) - Distance Education - 2016 CGPA - T.Y.B.C.A Semester VI & P131		
Year / Semester	T.Y.B.C.A / Semester VI		
Name of the Course & Course Code	Android Programming (CMP516)		
Name of the Student	SAYYED AFTAB AHMED ZULFIKAR AHMED		
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Regional Centre	Y.C.M.OPEN UNIVERSITY REGIONAL CENTER MUMBAI & 3139A		
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Name of the Study Centre & Code	CHANDRABHAN SHARMA COLLEGE OF ARTS, SCIENCE AND COMMERCE COLLEGE & 31468		
Date of Submission		No of Pages	5

General Instructions

1	Home assignment is useful for self-analysis of your preparation of the final examination and progress
2	Read all the questions and their marks in the home assignment carefully understand the Definitions and concepts properly
3	Read carefully the text, syllabus and summary related to home assignment.
4	Do not copy the points, matter from the text while writing home assignment mentions your one point /opinions whenever necessary
5	Your answer should include how you would apply the knowledge in real life situations
6	Use A4 size lined papers for writing home assignment by taking printout from this file only
7	Write each new answer on a different page.
8	Printed or typed answers are not allowed
9	The marks of Home assignment and corrective instructions will be sent after checking home assignment
10	Present the Home Assignment by following the instructions and rules. Read carefully all the instructions and rules before any correspondence or communication with the university
11	**For the programs under NEP 2020 Pattern each theory course of 4 credits its carry 30 marks, there are Two Home Assignments:** i) **Home Assignment 01 (HA01) is of Twenty marks** consisting of 2 Short Answer Questions carrying 5 marks each and 1 Long Answer Question of 10 marks. ii) **Home Assignment 02 (HA02) is of Ten marks** consisting of total 10 Objective Type Questions (MCQ) carrying 1 mark each.
12	**For each theory course of 2 credits, there is Only One Home Assignment of Fifteen Marks** consisting of 1 Short Answer Question of 5 marks and 1 Long Answer Question of 10 marks.
13	Student have to write the Home assignment 02 (MCQ) on separate page. Home Assignment 02 also to be uploaded along with Home assignment 01.

Student Signature

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Q.NO. Draw Android architecture and explain with all Components.

```
+-----------------------------------+
|          Applications             |
+-----------------------------------+
|      Application Framework        |
+-----------------------------------+
| Android Runtime & Libraries       |
+-----------------------------------+
|          Linux kernel             |
+-----------------------------------+
```

Explanation of Components.

1] Linux kernel
- Base layer of Android
- Manages hardware (CPU, Memory, drivers, Security)

2] Libraries
- Collection of C/C++ libraries
- provides features like media, graphics, database (SQ, lite), web browser

3] Android Runtime (ART)
- Executes Android applications
- Includes core libraries and manages app execution

4] Application framework
- provides Services like Activity Manager, window manager, content provider.
- Helps developers build apps easily

5] Applications
- Top layer (user apps)
- example :- whats app, youtube, chrome, etc.

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Q.NO.

Conclusion : Android architecture is layered, making it flexible secure, and easy for app development.

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Q.NO. What is CRUD operation in SQlite database? write the query for each operation.

CRUD stands for create, Read, update, and delete. These are the basic operation used to manage data in an SQLite database. in Android SQ Lite is used for storing local data.

1] Create (Insert Data)

Used to add new record into a table.
Query:

SQL:
SELECT INTO student (Id, name, age) VALUES (1, 'Aftab', 20);

2] Read (Retrieve Data)

used to fetch data from the table.
Query:

SQL:
SELECT * from Students;

3] update (Modify data)
Used to change existing record
Query:

SQL:-
Update Student SET age = 21 WHERE id = 1;

4] Delete (Remove data)
Used to delet record from the table
DELETE FROM student WHERE id = 1;
Conclusion:- CRUD operation are essential for handling data in SQLite, allowing, insertion

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Q.NO. Explain mobility management in GSM with call organisation and call terminatio

Mobility management in GSM is the process of tracking and Maintaining the location of a mobile user so that communication can continue while moving from one place to another. It ensure that users can make and receive calls without interruption.

function of Mobility Management
- Location updating of user
- paging (finding User for incoming call)
- Roaming support
- Authentication and Security

call origination process (outgoing call)
1] user dials a Number from mobile
2] Request is sent to nearest Base Station (BTS)
3] BTS forward it to mobile switching centre (MSC)
4] MSC checks user validity and Networks
5] call is connected to the destination user

call Termination process (Incoming call)
1] call request comes to MSC
2] MSC checks user's last known location
3] Network sends paging message to nearby BTS
4] Mobiles responds to the signal
5] call is connected to the user.

conclusion

Mobility management ensures smooth communication in GSM by tracking users and handling both outgoing and incoming call efficiently.

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Q.NO. Define and explain Android Operating System with at least five application

Android is an open-source mobile operating system developed by Google. It is based on the Linux kernel and is mainly used in smartphones, tablets, and other smart devices. Android provides a user-friendly interfaces, multitasking features, and support a wide range of applications.

Android allows developers to create app using languages like java and kotlin. It also provids features like touch interface, notification internet connectivity, GPS, camera, and multimedia support. Due to its flexibility and large app ecosystem, Android is the Most widely used mobile os in the world.

Applications of Android.

Some popular Android application are :-

- whatsapp - used for messaging and calling
- youtube - used for watching vedios
- Google maps - used for navigation and location
- Gmails - used for sending emails
- Goole chrome - used for browing the Internet.

Conclusion

Androids OS is powerful, flexible, and widely used, supporting millions of application that make daily tasks easier.

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